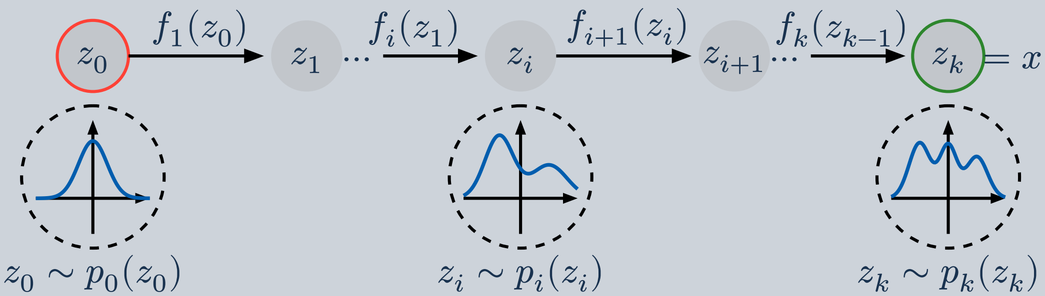
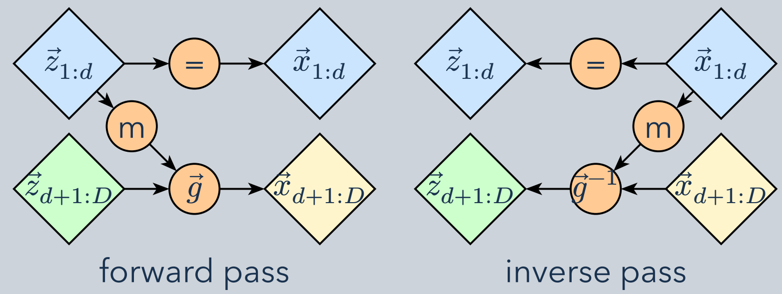
Build a complicated density by composing invertible transformations of a simple one. The Jacobian determinant keeps track of the volume change, so density stays normalized.

1 Compose invertible maps



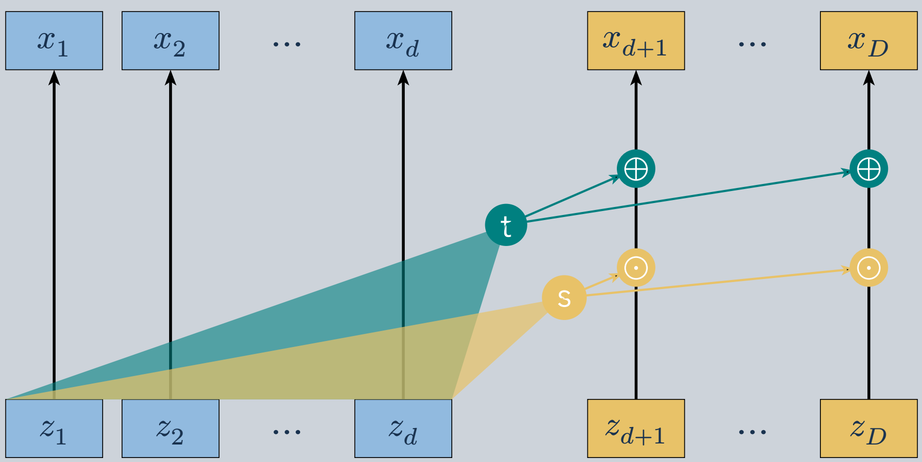
A base sample z_0 moves through $f_1, f_2, ..., f_K$ to become a data-space sample. Every step must have a computable inverse and density correction.

2 A coupling layer is reversible



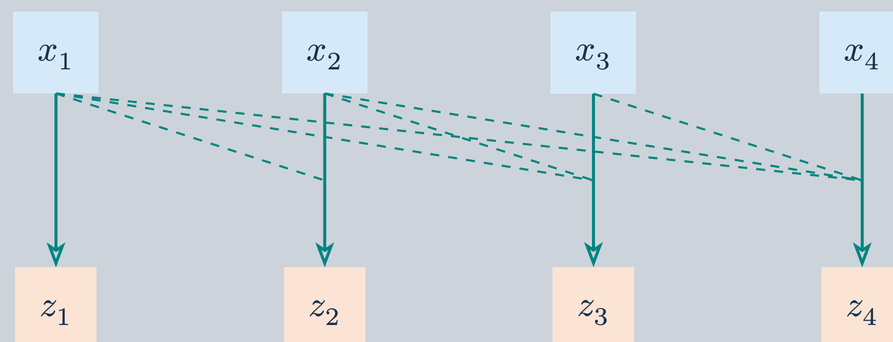
Leave one block unchanged; use it to condition an invertible map of the other block. The conditioner itself does not need to be invertible.

3 Affine coupling



Keep $x_A = z_A$ and transform $x_B = z_B \cdot \exp(s(z_A)) + t(z_A)$. Scale and shift depend only on the unchanged block; alternate which block is transformed.

4 Autoregressive conditioning



$$z_i = \left(x_i - \mu_{i(x_{<i})}\right) \exp\left(-s_{i(x_{<i})}\right)$$

For a masked autoregressive flow, coordinate i is conditioned on all earlier data coordinates. A masked network evaluates density in parallel; generation follows the ordering sequentially.

Change of variables: for $x = f(z)$, $\log p_{X(x)} = \log p_{Z(z)} - \log\left|\det J_{f(z)}\right|$. Triangular Jacobians make the determinant cheap: multiply the diagonal entries, or add their logarithms.